

WHATSAP REPLY BOT

1. PROJECT OVERVIEW (What the system will do)

A SaaS platform where businesses subscribe and connect their WhatsApp number, then the system automatically:

- replies to customers instantly
 - answers FAQs
 - sends price list / catalog
 - collects orders
 - sends order confirmation
 - optionally requests payment (M-Pesa)
 - provides admin dashboard for business owner
 - provides super admin dashboard for you
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2. TECHNICAL REQUIREMENTS (Hardware + Hosting)

Server Requirements

- VPS / Cloud Hosting (DigitalOcean / AWS / Contabo / Hetzner)
- Minimum:
 - 2 vCPU
 - 4GB RAM
 - 50GB SSD
- Ubuntu 22.04 recommended
- Nginx / Apache
- SSL Certificate (Let's Encrypt)

Software Requirements

- PHP 8.1+ (8.2 recommended)
 - Laravel 10/11
 - MySQL / MariaDB
 - Redis (optional but recommended)
 - Supervisor (for queue workers)
 - Cron job enabled
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3. API REQUIREMENTS (Very Important)

WhatsApp API Options

Required (Best):

- Meta WhatsApp Cloud API

You need:

- Meta Developer Account
 - Meta Business Account
 - WhatsApp Business Account (WABA)
 - Phone Number ID
 - Permanent Access Token
 - Webhook Verification Token
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AI Requirements

- OpenAI API Key (ChatGPT integration)
 - Model: GPT-4o-mini (cheap & fast)
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Payment Integration (Optional but powerful)

- Safaricom Daraja API (M-Pesa)
 - Consumer Key & Secret

- Shortcode
 - Passkey
 - Callback URL
-

4. CORE FUNCTIONAL REQUIREMENTS (Features)

A. Authentication & Roles

The system must support:

User Roles

1. **Super Admin** (You)
2. **Business Owner**
3. **Staff/Agents** (optional)

Requirements

- login/register
 - password reset
 - role-based access control
 - user profile settings
-

B. Business Management (Multi-Tenant SaaS)

The system must support multiple businesses.

Business Requirements

- business registration
- business profile setup:
 - name
 - category (salon, restaurant, shop, clinic etc.)
 - location

- working hours
 - delivery charges
 - social links
 - business can manage its own:
 - products
 - FAQs
 - automated messages
-

C. WhatsApp Connection Module

Each business must connect WhatsApp.

Requirements

- store WhatsApp credentials:
 - phone_number_id
 - access_token
 - verify_token
 - webhook URL per business or shared webhook
 - verify webhook automatically
 - enable/disable bot toggle
-

D. Webhook Message Receiver

System must be able to:

- receive incoming WhatsApp messages
 - log all messages
 - detect message type:
 - text
 - image
 - audio
 - document
 - location
 - save message in database
 - attach messages to a customer profile
-

E. Customer Management Module

The system should automatically create customers.

Requirements

- store customer details:
 - phone number
 - name (if available)
 - last seen
 - conversation status
 - customer segmentation:
 - new customers
 - repeat customers
-

F. Auto Reply Module (Rules Based)

Before AI, system must have automated replies:

Requirements

- greeting message (first message)
 - away message (outside working hours)
 - keyword replies:
 - price
 - location
 - opening hours
 - delivery
 - payment methods
 - quick menu reply:
 - “1. Prices”
 - “2. Order”
 - “3. Talk to human”
-

G. FAQ Management Module

Business must manage FAQs.

Requirements

- add FAQ question + answer
 - AI can use FAQ as knowledge base
 - ability to enable/disable FAQ entries
-

H. Product / Catalog Management Module

Business can manage their products.

Requirements

- add product name
- price
- description
- category
- product availability (in stock/out of stock)
- product image (optional)

Bot should respond:

- “Send catalog”
 - “Prices”
 - “Available products”
-

I. Order Management Module

System must allow customers to place orders via WhatsApp.

Requirements

- customer can order by:
 - product name
 - product number
 - menu selection
- bot collects:
 - items
 - quantities
 - delivery location
 - phone confirmation
- system generates order number
- order status tracking:

- pending
- confirmed
- dispatched
- delivered
- cancelled

Business dashboard should show orders list.

J. AI Chat Module (ChatGPT Integration)

The bot must answer naturally like a human.

Requirements

- AI responses based on:
 - business profile
 - products
 - FAQs
 - working hours
 - AI should never invent prices
 - AI must be limited with rules:
 - if message contains “price” → fetch from DB
 - if message contains “location” → fetch from DB
 - else use AI
-

K. Conversation Flow / Session State Module

The bot must remember where the customer is in conversation.

Requirements

- conversation steps:
 - greeting
 - menu
 - product selection
 - quantity
 - location
 - confirmation
- store conversation state in database
- reset conversation when finished

L. Human Agent / Manual Reply System

Business owner should take over chats.

Requirements

- toggle “human mode”
 - agent can respond manually from dashboard
 - bot pauses when human responds
 - chat inbox view like WhatsApp Web style
-

M. Notifications Module

Business must receive notifications.

Requirements

- notify business when:
 - new customer messages
 - new orders placed
 - notifications can be:
 - email
 - SMS (optional)
 - WhatsApp internal alert (optional)
-

N. Analytics & Reports Module

Business must see insights.

Requirements

- number of messages per day
- number of customers
- conversion rate (orders)
- most asked FAQs
- most ordered products
- sales reports (daily/weekly/monthly)

O. Subscription & Billing Module (SaaS monetization)

This is how you make money.

Requirements

- subscription plans:
 - Basic
 - AI Pro
 - Enterprise
 - plan limits:
 - number of messages/month
 - number of products
 - AI tokens usage
 - payment options:
 - M-Pesa Paybill/Till
 - Stripe (optional)
 - subscription expiry and renewal
 - auto disable bot if subscription expires
-

P. Audit Logs & Activity Tracking

For security and debugging.

Requirements

- log user actions:
 - who changed prices
 - who deleted orders
 - who updated FAQs
-

5. NON-FUNCTIONAL REQUIREMENTS (Quality Requirements)

Security Requirements

- HTTPS only
- API tokens encrypted in database
- prevent SQL injection / XSS
- secure webhook verification
- role-based permissions
- rate limiting (prevent spam)

Performance Requirements

- use Laravel Queue for message processing
- store logs efficiently
- cache business FAQ/product data
- support 1000+ messages/day without downtime

Reliability Requirements

- bot must retry sending message if API fails
- webhook must always return 200 quickly
- backup database daily

Scalability Requirements

- multi-business support
- scalable database structure
- separate queue workers for AI



6. DATABASE REQUIREMENTS (Core Tables Needed)

Main tables

- users
- roles / permissions
- businesses
- business_settings

- whatsapp_accounts

Customer & chat tables

- customers
- conversations
- messages

Product & order tables

- products
- categories
- orders
- order_items

FAQ & automation tables

- faqs
- auto_replies
- bot_rules

Subscription tables

- plans
- subscriptions
- payments
- invoices

Logs

- activity_logs
- webhook_logs
- api_request_logs

7. UI REQUIREMENTS (Dashboard Screens)

Business Owner Dashboard

- Overview page
- WhatsApp setup page
- Chat inbox page
- Customers page
- Orders page
- Products management page
- FAQ management page
- Settings page
- Subscription billing page
- Reports page

Super Admin Dashboard (You)

- all businesses list
 - manage subscriptions
 - monitor usage
 - system logs
 - manage plans/pricing
 - analytics for whole platform
-

8. SYSTEM MODULE REQUIREMENTS (Backend Structure)

Modules / Services

- WhatsAppWebhookService
 - WhatsAppMessageSenderService
 - AIReplyService
 - FAQService
 - OrderService
 - ConversationStateService
 - SubscriptionService
 - NotificationService
 - LoggingService
-

9. DEPLOYMENT REQUIREMENTS

- domain name (example: bot.afrispark.com)
 - SSL installed
 - cron job:
 - schedule:run
 - queue worker:
 - Supervisor config
 - database backups
 - logging + monitoring
-

10. LEGAL / POLICY REQUIREMENTS (Meta rules)

Meta requires:

- Privacy policy page
 - Terms & Conditions page
 - Data deletion request page
 - User consent handling
 - Message template approval (for business-initiated messages)
-

11. OPTIONAL ADVANCED FEATURES (Upgrade Level)

 These make your system premium:

- voice note transcription (AI converts voice to text)
 - multilingual support (English/Swahili)
 - AI product recommendations
 - customer loyalty points
 - integration with Google Sheets
 - integration with POS system
 - chatbot training panel (custom knowledge base PDF upload)
-

✓ 12. REQUIREMENTS FOR MVP (Minimum Product to Sell Fast)

If you want a version you can sell within 2 weeks:

MVP Must Have:

- ✓ WhatsApp webhook receive + reply
 - ✓ FAQ replies
 - ✓ price list reply
 - ✓ product list
 - ✓ order capturing (simple)
 - ✓ dashboard to view orders/messages
 - ✓ AI for unknown questions
 - ✓ subscription expiry system
-

🚀 FINAL LIST SUMMARY (What you need to start)

To build the full system you need:

Technical

- Laravel + MySQL + VPS + SSL

APIs

- Meta WhatsApp Cloud API
- OpenAI API
- (Optional) M-Pesa Daraja API

Core Features

- WhatsApp webhook
- auto replies
- FAQ + catalog
- AI responses
- order system

- dashboard
- subscription billing

If you want, I can now write for you a **full SRS document (Software Requirements Specification)** format, like a real university/project document (Chapter 1–5 style) so you can even use it for proposals and pitching.

Perfect 🔥 Let's make your project look like a full serious system.

Below is a **complete Database Design (ERD Tables + Relationships)** and also **DFD Level 0 and Level 1** for the **AI WhatsApp Business Reply Bot & Order Management System**.

✅ PART A: DATABASE DESIGN (ERD TABLES + RELATIONSHIPS)

1. USERS & ACCESS CONTROL TABLES

1.1 users

Stores system users (Super Admin, Business Owners, Agents).

Field	Type	Description
id	BIGINT PK	Primary Key
name	VARCHAR	User full name

email	VARCHAR UNIQUE	Email address
phone	VARCHAR	Phone number
password	VARCHAR	Encrypted password
role_id	BIGINT FK	Role reference
business_id	BIGINT FK NULL	Belongs to a business
status	ENUM(active,inactive)	Account status
created_at	TIMESTAMP	Date created
updated_at	TIMESTAMP	Date updated

 Relationship:

- **users.role_id → roles.id**
- **users.business_id → businesses.id**

1.2 roles

Stores roles.

Field	Type
-------	------

id BIGINT PK

name VARCHAR

Examples:

- Super Admin
- Business Owner
- Agent

 Relationship:

- roles has many users

1.3 permissions (*Optional but recommended*)

Field	Type
-------	------

id	BIGINT PK
----	-----------

name	VARCHAR
------	---------

1.4 role_permissions (*Pivot table*)

Field	Type
-------	------

id BIGINT PK

role_id BIGINT FK

permission_id BIGINT FK

 Relationship:

- role_permissions.role_id → roles.id
- role_permissions.permission_id → permissions.id

2. BUSINESS MANAGEMENT TABLES

2.1 businesses

Stores registered businesses.

Field	Type	Description
id	BIGINT PK	Primary key
name	VARCHAR	Business name
category	VARCHAR	Salon, restaurant, shop etc

email	VARCHAR	Business email
phone	VARCHAR	Business contact
location	TEXT	Business location
description	TEXT	Business description
opening_hours	TEXT	Opening and closing hours
delivery_info	TEXT	Delivery charges & rules
status	ENUM(active,inactive)	Business status
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

Relationship:

- businesses has many users
- businesses has one whatsapp_accounts
- businesses has many products
- businesses has many faqs
- businesses has many customers
- businesses has many orders

2.2 business_settings

Stores business bot configuration.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
greeting_message	TEXT
away_message	TEXT
fallback_message	TEXT
bot_enabled	BOOLEAN
ai_enabled	BOOLEAN
human_takeover	BOOLEAN
language	VARCHAR
created_at	TIMESTAMP
updated_at	TIMESTAMP

 Relationship:

- `business_settings.business_id` → `businesses.id`
-
-

3. WHATSAPP INTEGRATION TABLES

3.1 whatsapp_accounts

Stores WhatsApp API details per business.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
phone_number_id	VARCHAR
whatsapp_business_account_id	VARCHAR
access_token	TEXT (encrypted)
verify_token	VARCHAR
webhook_url	TEXT

status	ENUM(active,inactive)
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created_at	TIMESTAMP
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updated_at	TIMESTAMP
------------	-----------

 Relationship:

- whatsapp_accounts.business_id → businesses.id
-

3.2 webhook_logs

Stores webhook payload logs for debugging.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
payload	LONGTEXT
event_type	VARCHAR
created_at	TIMESTAMP

 Relationship:

- webhook_logs.business_id → businesses.id
-
-

4. CUSTOMER & CHAT TABLES

4.1 customers

Stores customers who message the business.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
name	VARCHAR NULL
phone	VARCHAR
last_seen	TIMESTAMP
created_at	TIMESTAMP
updated_at	TIMESTAMP

 Relationship:

- customers.business_id → businesses.id

- customers has many messages
 - customers has many orders
 - customers has one conversation_state
-

4.2 conversations

Tracks conversation state for order flow.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
customer_id	BIGINT FK
current_step	VARCHAR
session_data	JSON
is_active	BOOLEAN
last_message_ at	TIMESTAMP
created_at	TIMESTAMP

updated_at TIMESTAMP

 Relationship:

- conversations.customer_id → customers.id
- conversations.business_id → businesses.id

4.3 messages

Stores all incoming/outgoing WhatsApp messages.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
customer_id	BIGINT FK
message_id	VARCHAR
direction	ENUM(incoming,outgoing)
message_type	ENUM(text,image,audio,document,location)
content	LONGTEXT

status ENUM(sent,failed,received)

created_at TIMESTAMP

📌 Relationship:

- messages.business_id → businesses.id
 - messages.customer_id → customers.id
-
-

✅ 5. FAQ & AUTO REPLY TABLES

5.1 faqs

Stores FAQ questions and answers.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
question	TEXT
answer	TEXT

keywords TEXT

is_active BOOLEAN

created_at TIMESTAMP

updated_at TIMESTAMP

 Relationship:

- faqs.business_id → businesses.id

5.2 auto_replies

Stores rule-based replies.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
trigger_keyword	VARCHAR
reply_message	TEXT

match_type ENUM(exact,contains,regex)

is_active BOOLEAN

created_at TIMESTAMP

updated_at TIMESTAMP

📌 Relationship:

- auto_replies.business_id → businesses.id

✅ 6. PRODUCTS & CATALOG TABLES

6.1 categories

Product categories.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
name	VARCHAR

created_at TIMESTAMP

 Relationship:

- categories.business_id → businesses.id
- categories has many products

6.2 products

Stores product/services list.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
category_id	BIGINT FK
name	VARCHAR
description	TEXT
price	DECIMAL
stock_status	ENUM(in_stock,out_of_stock)

image VARCHAR NULL

is_active BOOLEAN

created_at TIMESTAMP

updated_at TIMESTAMP

 Relationship:

- products.business_id → businesses.id
 - products.category_id → categories.id
-
-

7. ORDER MANAGEMENT TABLES

7.1 orders

Stores customer orders.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
customer_id	BIGINT FK

order_number	VARCHAR UNIQUE
status	ENUM(pending,confirmed,dispatched,delivered,cancelled)
delivery_location	TEXT
total_amount	DECIMAL
payment_status	ENUM(unpaid,paid,partial)
notes	TEXT
created_at	TIMESTAMP
updated_at	TIMESTAMP

Relationship:

- orders.business_id → businesses.id
- orders.customer_id → customers.id
- orders has many order_items

7.2 order_items

Stores order products.

Field	Type
-------	------

id	BIGINT PK
order_id	BIGINT FK
product_id	BIGINT FK NULL
product_name	VARCHAR
quantity	INT
unit_price	DECIMAL
subtotal	DECIMAL

📌 Relationship:

- order_items.order_id → orders.id
 - order_items.product_id → products.id
-
-

✅ 8. PAYMENT & SUBSCRIPTION TABLES (SaaS Monetization)

8.1 plans

Subscription plans.

Field	Type
id	BIGINT PK
name	VARCHAR
price	DECIMAL
duration_days	INT
message_limit	INT
ai_enabled	BOOLEAN
created_at	TIMESTAMP

 Relationship:

- plans has many subscriptions

8.2 subscriptions

Business subscriptions.

Field	Type
-------	------

id	BIGINT PK
business_id	BIGINT FK
plan_id	BIGINT FK
start_date	DATE
end_date	DATE
status	ENUM(active,expired,cancelled)
created_at	TIMESTAMP

 Relationship:

- subscriptions.business_id → businesses.id
- subscriptions.plan_id → plans.id

8.3 payments

Stores payments made by business owners.

Field	Type
id	BIGINT PK

business_id	BIGINT FK
subscription_id	BIGINT FK
amount	DECIMAL
method	ENUM(mpesa,stripe,cash)
transaction_code	VARCHAR
status	ENUM(pending,paid,failed)
paid_at	TIMESTAMP
created_at	TIMESTAMP

📌 Relationship:

- payments.business_id → businesses.id
- payments.subscription_id → subscriptions.id

9. AI REQUEST LOG TABLES

9.1 ai_logs

Stores AI prompts and responses.

Field	Type
id	BIGINT PK
business_id	BIGINT FK
customer_id	BIGINT FK
prompt	LONGTEXT
response	LONGTEXT
tokens_used	INT
created_at	TIMESTAMP

📌 Relationship:

- ai_logs.business_id → businesses.id
- ai_logs.customer_id → customers.id



10. SYSTEM LOGS TABLES

10.1 activity_logs

Tracks actions by business owner/staff.

Field	Type
id	BIGINT PK
user_id	BIGINT FK
business_id	BIGINT FK
action	VARCHAR
description	TEXT
created_at	TIMESTAMP

📌 Relationship:

- activity_logs.user_id → users.id
 - activity_logs.business_id → businesses.id
-
-

✅ ERD RELATIONSHIP SUMMARY (Quick View)

Main Relationships

- **roles 1 → many users**
- **businesses 1 → many users**
- **businesses 1 → 1 whatsapp_accounts**

- **businesses 1 → 1 business_settings**
- **businesses 1 → many customers**
- **customers 1 → many messages**
- **customers 1 → many orders**
- **orders 1 → many order_items**
- **businesses 1 → many products**
- **categories 1 → many products**
- **businesses 1 → many faqs**
- **businesses 1 → many subscriptions**
- **subscriptions 1 → many payments**
- **businesses 1 → many ai_logs**

That is your full ERD structure.



PART B: DATA FLOW DIAGRAM (DFD)

Now let's do DFD properly.



DFD LEVEL 0 (Context Diagram)

External Entities

1. Customer (WhatsApp User)
 2. Business Owner
 3. Meta WhatsApp Cloud API
 4. OpenAI API
 5. Payment Provider (M-Pesa Daraja API)
 6. Super Admin
-

DFD Level 0 Flow

Customer → System

- sends message inquiry
- requests catalog/prices

- places order

System → Customer

- auto reply response
- price list
- order confirmation

System ↔ Meta WhatsApp Cloud API

- receives incoming message webhook
- sends outgoing WhatsApp replies

System ↔ OpenAI API

- sends prompt request
- receives AI response

System ↔ M-Pesa API

- sends STK push request
- receives payment confirmation callback

Business Owner ↔ System

- manages products
- manages FAQs
- views messages and orders
- responds manually

Super Admin ↔ System

- manages subscriptions/plans
- monitors businesses and logs

DFD Level 0 Diagram Representation (Text)

Customer (WhatsApp)

|

| Message/Order Request

v

[AI WhatsApp Bot System] <-----> Meta WhatsApp Cloud API

|

| AI Query

v

OpenAI API

|

| Payment Request

v

M-Pesa API

Business Owner <-----> [AI WhatsApp Bot System] <-----> Super Admin



DFD LEVEL 1 (Detailed Diagram)

Now we break the system into internal processes.

Processes in Level 1

- 1.0 User Authentication & Access Control
- 2.0 WhatsApp Message Handling
- 3.0 FAQ & Auto Reply Processing
- 4.0 AI Response Generation
- 5.0 Product & Price List Processing
- 6.0 Order Management Processing
- 7.0 Subscription & Billing Processing
- 8.0 Reporting & Analytics

Data Stores (Database)

D1 Users Table
D2 Businesses Table
D3 WhatsApp Accounts Table
D4 Customers Table
D5 Messages Table
D6 FAQs Table
D7 Products Table
D8 Orders Table
D9 Subscription Table
D10 AI Logs Table

DFD LEVEL 1 FLOWS (Detailed)

1.0 Authentication Process

Input:

- login credentials from Business Owner / Super Admin

Output:

- verified access session
- dashboard access

Data Store Used:

- D1 Users Table
 - D2 Businesses Table
-

2.0 WhatsApp Message Handling

Input:

- customer message from Meta WhatsApp API

Output:

- stored incoming message
- message forwarded to reply processing

Data Store Used:

- D4 Customers Table
 - D5 Messages Table
-

3.0 FAQ & Auto Reply Processing

Input:

- incoming customer message

Output:

- reply message if keyword matches FAQ/auto reply rules

Data Store Used:

- D6 FAQs Table
 - Auto Replies Table
-

4.0 AI Response Generation

Input:

- customer message (if not matched in FAQ)

- business data (products, FAQs, settings)

Output:

- AI-generated response

External Entity:

- OpenAI API

Data Store Used:

- D10 AI Logs Table
 - D2 Businesses Table
 - D7 Products Table
 - D6 FAQs Table
-

5.0 Product & Price List Processing

Input:

- customer request: "prices", "catalog"

Output:

- formatted product list + price list message

Data Store Used:

- D7 Products Table
 - Categories Table
-

6.0 Order Management Processing

Input:

- customer order request

Output:

- order created
- order confirmation message

Data Store Used:

- D8 Orders Table
 - D8.1 Order Items Table
 - D4 Customers Table
-

7.0 Subscription & Billing Processing

Input:

- subscription request from business owner
- payment confirmation from M-Pesa

Output:

- subscription activated
- invoice/payment record created

External Entity:

- M-Pesa API

Data Store Used:

- D9 Subscription Table
 - Payments Table
-

8.0 Reporting & Analytics

Input:

- messages and orders history

Output:

- charts and reports dashboard

Data Store Used:

- D5 Messages Table
 - D8 Orders Table
 - D4 Customers Table
-
-



DFD LEVEL 1 (Text Diagram)

Customer

|

| WhatsApp Message

v

Meta WhatsApp API

|

| Webhook

v

(2.0 WhatsApp Message Handling) ---> D4 Customers

|

D5 Messages

|

v

(3.0 FAQ & Auto Reply Processing) ---> D6 FAQs

|

| if no match

v

(4.0 AI Response Generation) <-----> OpenAI API

| D10 AI Logs

|

v

(5.0 Product & Price List Processing) ---> D7 Products

|

v

(6.0 Order Management Processing) ---> D8 Orders + Order Items

|

v

Meta WhatsApp API

|

| Reply Message

v

Customer

Business Owner ---> (1.0 Authentication) ---> D1 Users

Business Owner ---> Manage Products/FAQs ---> D7 Products / D6 FAQs

Business Owner ---> View Orders -----> D8 Orders

Business Owner ---> Subscription -----> (7.0 Subscription & Billing) <-----> M-Pesa API

Super Admin -----> Manage System -----> D2 Businesses / D9 Subscriptions

OPTIONAL DFD LEVEL 1 PROCESSES (Extra Details)

If you want to make it even more academic, you can include:

- 2.1 Validate Webhook Token
- 2.2 Save Incoming Message
- 2.3 Detect Message Type
- 6.1 Create Order
- 6.2 Update Order Status
- 6.3 Send Order Notification to Business Owner
- 4.1 Build AI Prompt
- 4.2 Fetch Business Context
- 4.3 Generate AI Response

This makes your DFD look extremely professional.

FINAL OUTPUT SUMMARY (What you now have)

-  Full ERD database tables
 -  Table fields + keys + relationships
 -  DFD Level 0 (Context Diagram)
 -  DFD Level 1 (Detailed system flow)
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If you want, I can now generate for you the **actual ERD diagram format** (like MySQL Workbench style explanation) and also write the **SQL migration script structure** for Laravel.